

Mole Conversion Worksheet

$ \begin{array}{c} \div M \\ \xrightarrow{\hspace{1cm}} \\ \text{m (mass in grams)} \\ \xleftarrow{\hspace{1cm}} \\ \times M \end{array} $	$ \begin{array}{c} \times N \\ \xrightarrow{\hspace{1cm}} \\ \text{moles} \\ \xleftarrow{\hspace{1cm}} \\ \div N \end{array} $	$ \begin{array}{c} \times \text{subscripts} \\ \xrightarrow{\hspace{1cm}} \\ \text{\# molecules or units} \\ \xleftarrow{\hspace{1cm}} \\ \div \text{subscripts} \end{array} $	$ \begin{array}{c} \text{\# atoms} \end{array} $
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9.8 g H ₂ SO ₄			hydrogen
	3.0 mol NaCl		1.8 x 10 ²⁴ chlorine
		1.5 x 10 ²⁷ molecules SO ₂	oxygen
15.96 g CuSO ₄ •5H ₂ O			hydrogen
	4.7 x 10 ⁻² mol KMnO ₄		oxygen
		7.8 x 10 ²⁶ molecules C ₆ H ₆	carbon
g H ₂ O			1.2 x 10 ²³ oxygen
	9.3 x 10 ⁻⁵ mol Pb(NO ₃) ₂		nitrogen
		5.6 x 10 ³⁰ molecules C ₂ H ₅ OH	hydrogen
g CaCl ₂			1.1 x 10 ²² chlorine
	2.3 mol N ₂ O ₄		5.6 x 10 ²⁴ oxygen